

TCRR Phase 1 — Benchmarking Report

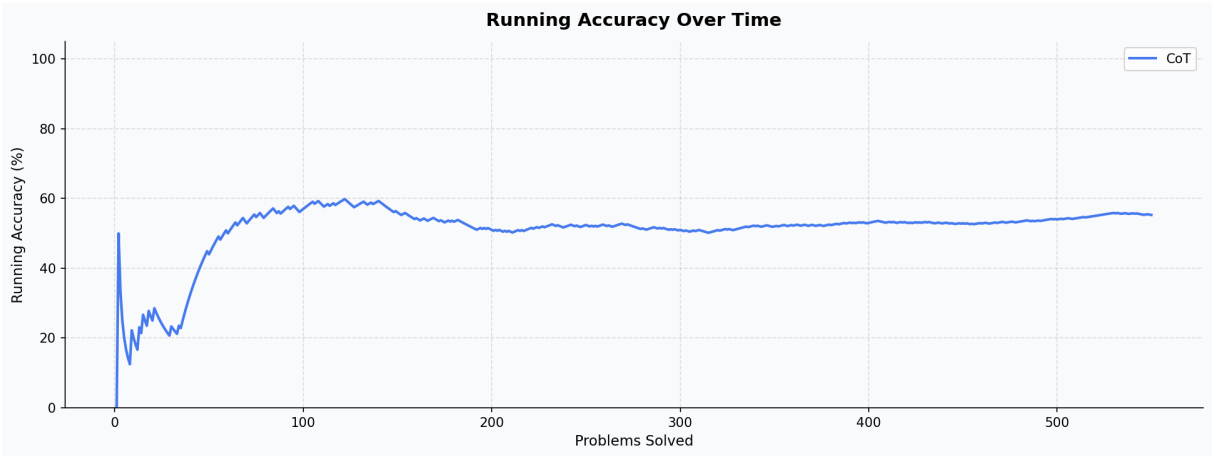
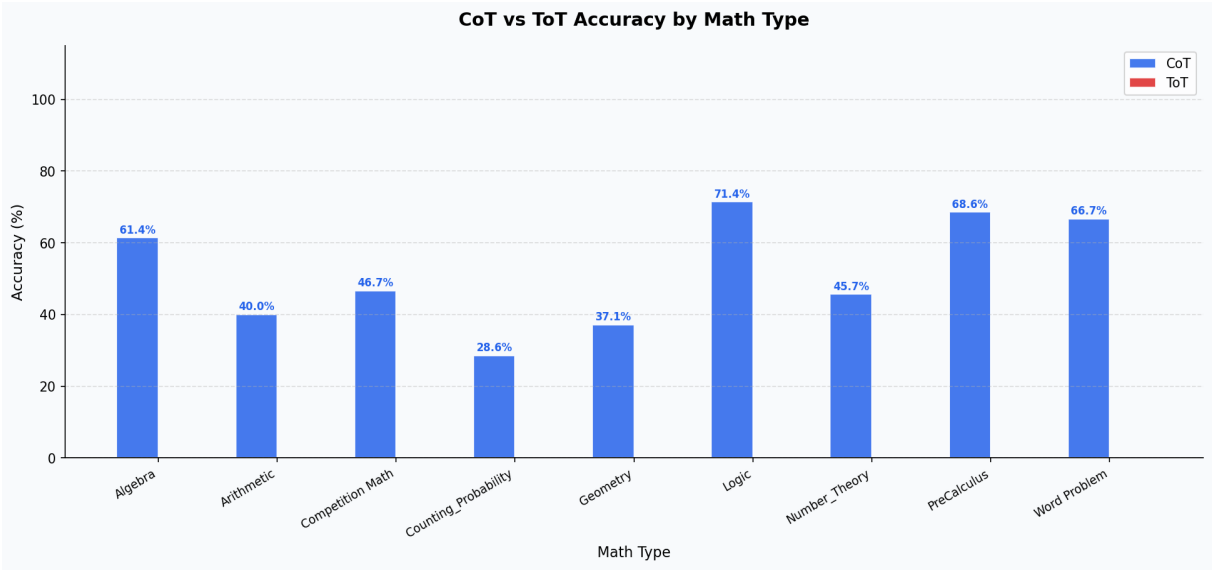
Type-Aware Cost-Efficient Reasoning Router

Questions: 550 | Generated: 2026-03-12 14:32:02 | Model: llama-3.3-70b-versatile (Groq)

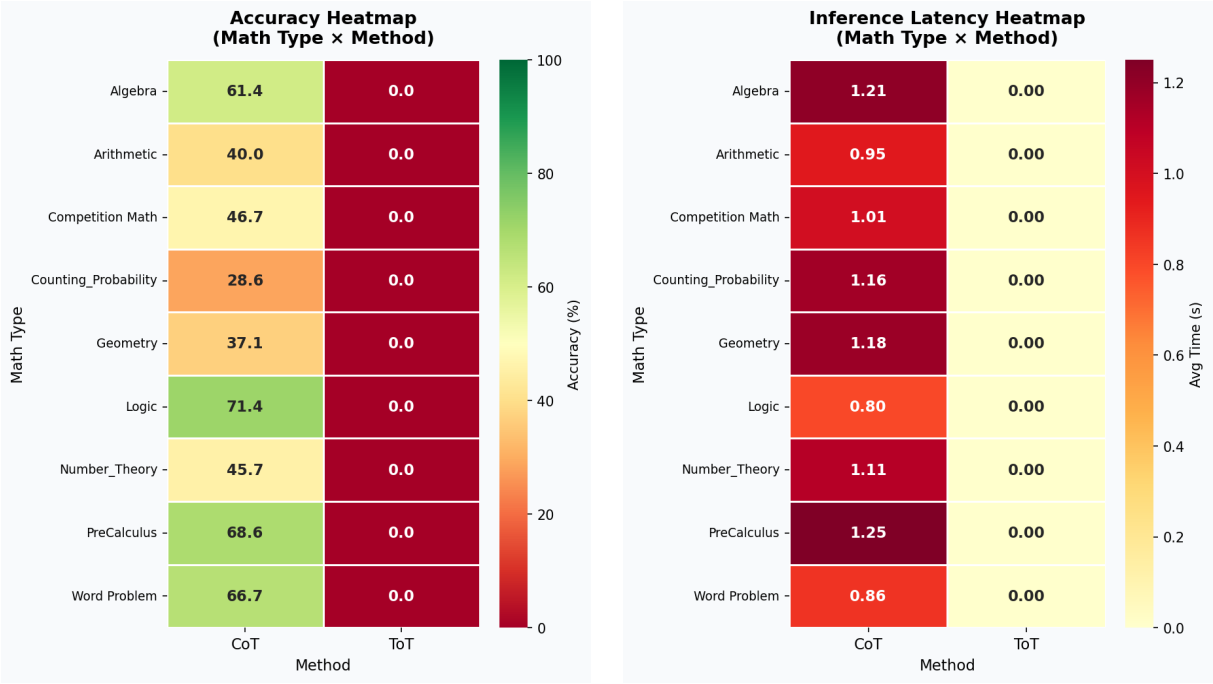
1. Per-Category Summary

Category	n	CoT Acc%	ToT Acc%	CoT Lat(s)	ToT Lat(s)	CoT Tokens	ToT Tokens	CoT Steps	ToT Steps	Winner
Algebra	70	61.4	—	1.21	—	840.0	—	6.5	—	—
Arithmetic	35	40.0	—	0.95	—	654.0	—	5.3	—	—
Competition Math	105	46.7	—	1.01	—	683.0	—	6.2	—	—
Counting_Probability	35	28.6	—	1.16	—	809.0	—	4.5	—	—
Geometry	35	37.1	—	1.18	—	914.0	—	8.0	—	—
Logic	35	71.4	—	0.8	—	458.0	—	3.9	—	—
Number_Theory	35	45.7	—	1.11	—	744.0	—	7.1	—	—
PreCalculus	35	68.6	—	1.25	—	956.0	—	6.6	—	—
Word Problem	165	66.7	—	0.86	—	579.0	—	2.4	—	—
Overall	550/0	55.3	—	1.01	—	700.0	—	—	—	—

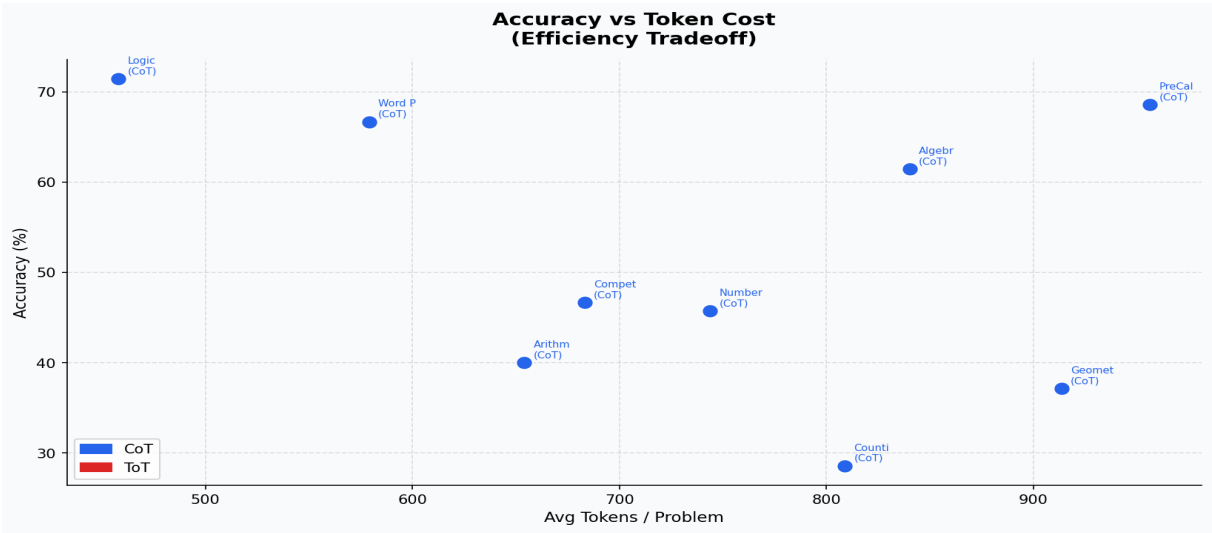
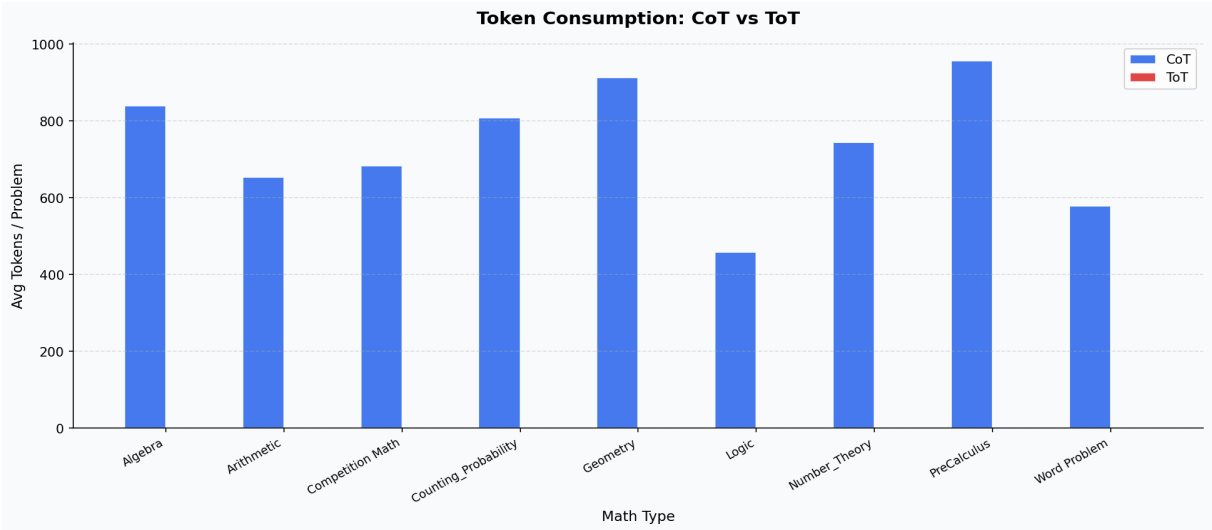
2. Accuracy Analysis



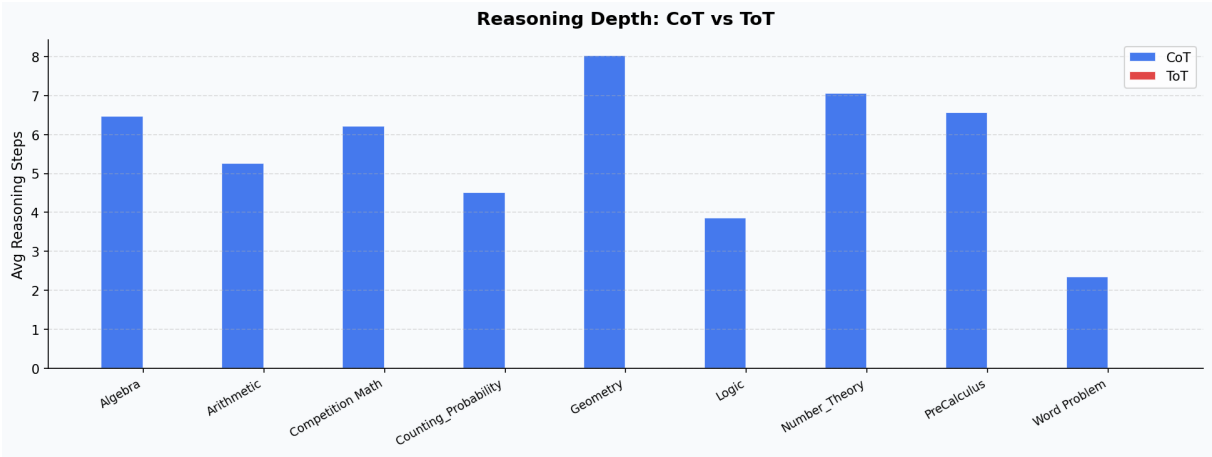
3. Heatmaps



4. Token Cost & Efficiency



5. Reasoning Depth & Strategy



6. Router Recommendation (Phase 2)

Math Type	CoT Acc%	ToT Acc%	Recommended	Token Δ	Reason
Algebra	—	—	Need data	—	—
Arithmetic	—	—	Need data	—	—
Competition Math	—	—	Need data	—	—
Counting_Probability	—	—	Need data	—	—
Geometry	—	—	Need data	—	—
Logic	—	—	Need data	—	—
Number_Theory	—	—	Need data	—	—
PreCalculus	—	—	Need data	—	—
Word Problem	—	—	Need data	—	—